

Incident Detection and Analysis

Incident Investigation in Splunk

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Incident Detection and Analysis

1 Context Overview

A SIEM alert triggered on a successful sign-in to a Nexus Financial user account from an IP address that was never observed in the environment before. All Nexus Financial employees work from the London office (IP: 197.32.45.112) and are expected to sign in from the corporate network. This sign-in came from outside the United Kingdom. Laura Chen (Finance Manager) has confirmed she did not initiate this sign-in. Initial triage indicates this is a true positive requiring full IR investigation.

- Alert Name: Anomalous Sign-in Detected
- Affected Account: l.chen@nexusfinancial.thm
- Corporate IP: 197.32.45.112
- Severity: High

Incident Detection and Analysis

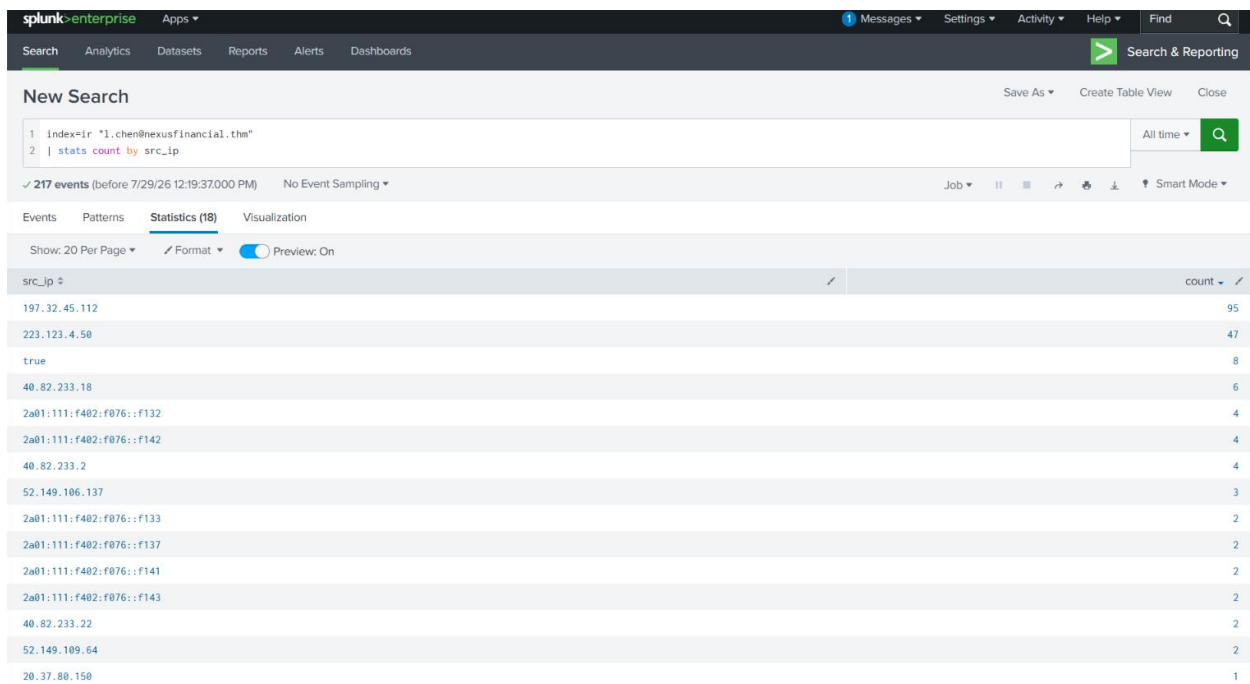
2 Hunting

2.1 IoC Detection

The following query was used to see the list of IP addresses making requests related to the email.

```
index=ir "l.chen@nexusfinancial.thm"
```

```
/ stats count by src_ip
```



src_ip	count
197.32.45.112	95
223.123.4.50	47
true	8
40.82.233.18	6
2a01:111:f402:f076::f132	4
2a01:111:f402:f076::f142	4
40.82.233.2	4
52.149.106.137	3
2a01:111:f402:f076::f133	2
2a01:111:f402:f076::f137	2
2a01:111:f402:f076::f141	2
2a01:111:f402:f076::f143	2
40.82.233.22	2
52.149.109.64	2
20.37.80.150	1

Figure 1 Query 1 result

The table shows that two IP addresses had made more requests than others. 197.32.45.112 is the corporate IP address. So, 223.123.4.50 could be the attacker's IP address.

The location of this IP address was investigated with the following query:

```
index=ir "l.chen@nexusfinancial.thm" src_ip="223.123.4.50"
```

```
/ table _time, location.city
```

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The screenshot shows the Splunk Enterprise Search interface. At the top, there's a navigation bar with 'splunk enterprise' logo and various menu items like 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'New Search' section with a search bar containing two queries: 1. `index=ir "l.chen@nexusfinancial.thm" src_ip="223.123.4.50"` and 2. `| table _time, location.city`. The search results show 47 events. The table has two columns: `_time` and `location.city`. The results show a list of timestamps and the location 'Amsterdam' for each event.

_time	location.city
2026-03-30 16:41:28	Amsterdam
2026-03-30 16:41:30	Amsterdam
2026-03-30 16:41:32	Amsterdam
2026-03-30 16:41:40	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:42:03	Amsterdam
2026-03-30 16:47:07	Amsterdam
2026-03-30 16:47:17	Amsterdam
2026-03-30 16:47:17	Amsterdam
2026-03-30 16:47:17	Amsterdam
2026-03-30 16:47:17	Amsterdam
2026-03-30 16:47:17	Amsterdam

Figure 2 Query II result

The IP address originated from Amsterdam. This confirms that since it originated from outside the organization, it is the adversary.

The next step was to detect when the attacker successfully signed on to Laura's account. Following query was used:

```
index=ir "l.chen@nexusfinancial.thm" src_ip="223.123.4.50" eventtype="azure_aad_signin"
action=success

| table _time
```

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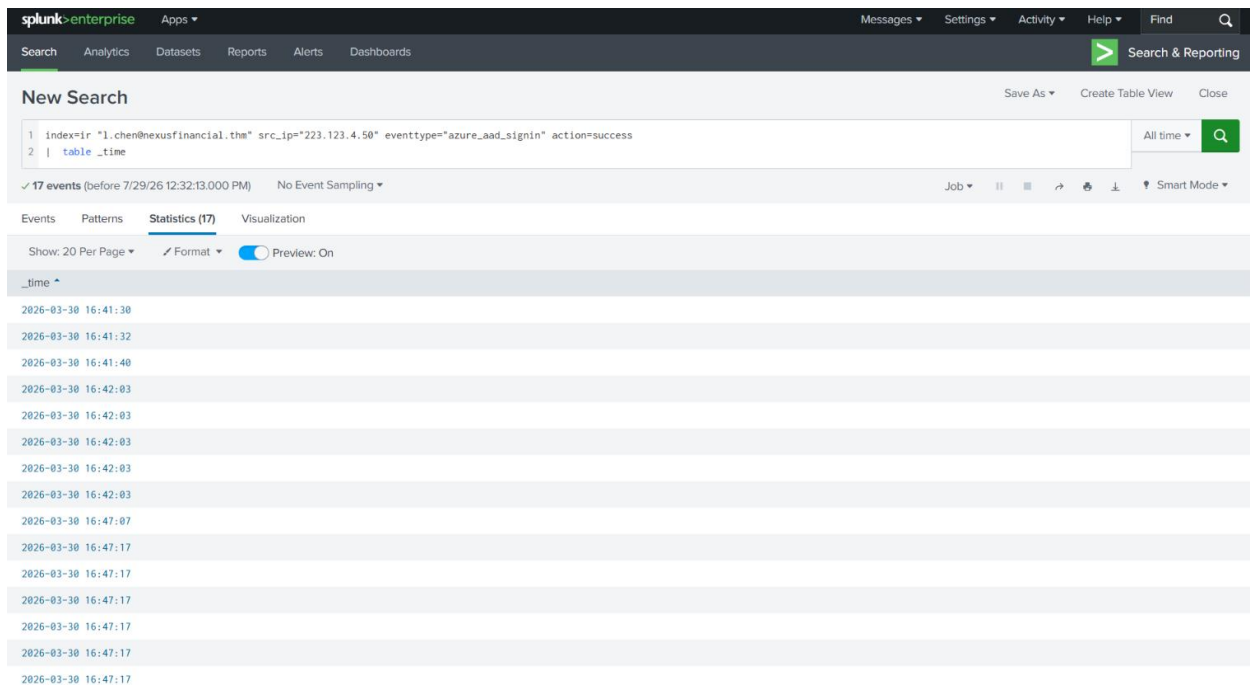


Figure 3 Query III result

The result indicated that the first login was successful at 2026-03-30 16:41:30.

The next step is to investigate the email sent to Laura before the login attempt. Following query was used:

`index=ir recipient="l.chen@nexusfinancial.thm"`

`| table _time, SenderAddress, Status, Subject`

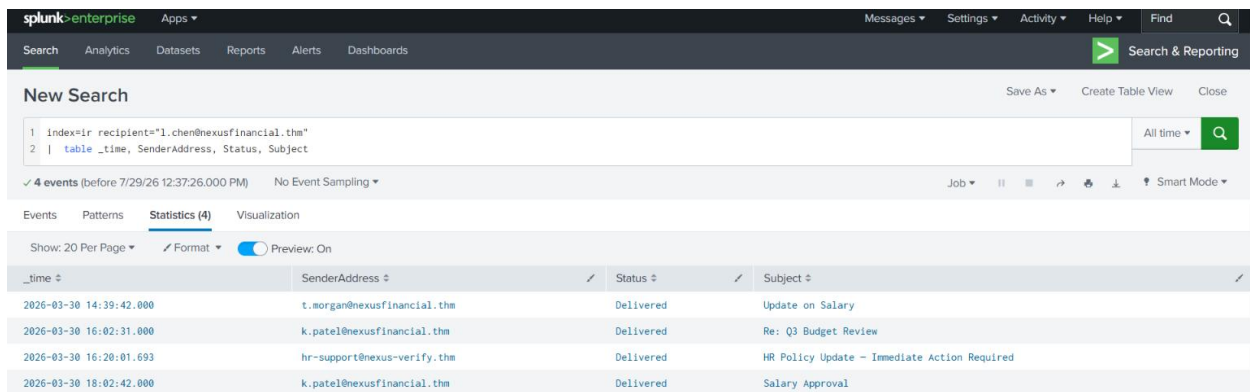


Figure 4 Query IV result

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The result shows that Laura has received four emails. Looking at the subject, the third email looks suspicious because the subject of the email indicates urgency, a common tactic used for phishing emails.

Following query was used to investigate the email:

index=ir recipient="l.chen@nexusfinancial.thm" SenderAddress="hr-support@nexus-verify.thm"

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk-enterprise' and various menu items like 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'New Search' section where the query `index=ir recipient="l.chen@nexusfinancial.thm" SenderAddress="hr-support@nexus-verify.thm"` is entered. The search results show 1 event. The event details are displayed in a table-like format with fields for Time, Event, and various email metadata.

Time	Event
3/30/26 4:20:01.693 PM	<pre>[{"FromIP": "2a00:1450:4864:20::234", "MessageId": "<CAA0ajKsy_vZw4BDjaiqNLT7HCpY1Kjxii+LK9rb1BW65PSvNxx@mail.gmail.com>", "MessageTraceId": "be10618b-d445-472f-66b0-08de8e7831dc", "Received": "2026-03-30T16:20:01.693Z", "RecipientAddress": "l.chen@nexusfinancial.thm", "SenderAddress": "hr-support@nexus-verify.thm", "Size": 53776, "Status": "Delivered", "Subject": "HR Policy Update - Immediate Action Required", "ToIP": "2a00:1450:4864:20::234"}]</pre> Show as raw text <pre>FromIP = 2a00:1450:4864:20::234 MessageId = <CAA0ajKsy_vZw4BDjaiqNLT7HCpY1Kjxii+LK9rb1BW65PSvNxx@mail.gmail.com> MessageTraceId = be10618b-d445-472f-66b0-08de8e7831dc Received = 2026-03-30T16:20:01.693Z RecipientAddress = l.chen@nexusfinancial.thm SenderAddress = hr-support@nexus-verify.thm Size = 53776 Status = Delivered Subject = HR Policy Update - Immediate Action Required ToIP = 2a00:1450:4864:20::234 action = delivered date_hour = 16 date_mday = 30 date_minute = 20 date_month = march date_second = 1 date_vday = monday date_year = 2026 date_zone = 0 dest = eventtype = o365_reporting_message host = ce-splunk index = ir internal_message_id = be10618b-d445-472f-66b0-08de8e7831dc linecount = 1</pre>

Figure 5 Query V result

The following information were gathered from this log:

1. The domain of the sender was nexus-verify.thm.
2. The message size was 53776.
3. The message ID was `<CAA0ajKsy_vZw4BDjaiqNLT7HCpY1Kjxii+LK9rb1BW65PSvNxx@mail.gmail.com>`

Incident Detection and Analysis

2.2 Analysis

The first step of analysis was to identify where the attackers attempted to sign in. Following query was used for this analysis:

```
index=ir src_ip="223.123.4.50" eventtype="azure_aad_signin" action="success"  
  
| stats count by user_id
```

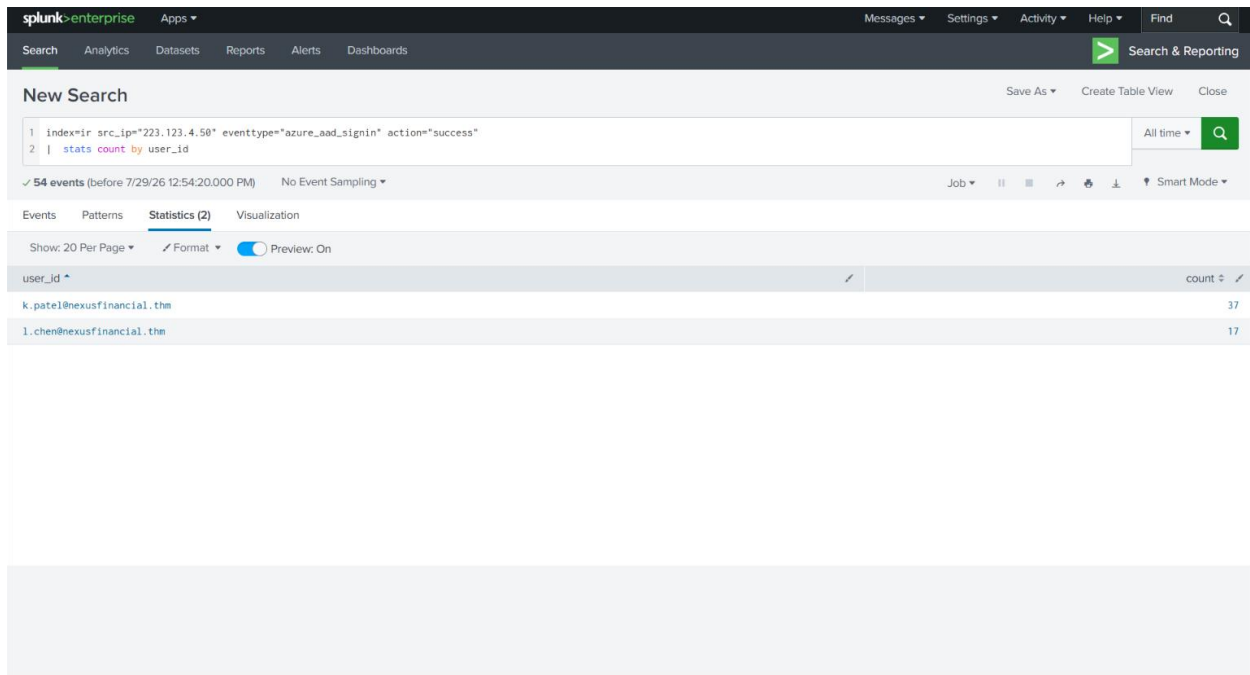


Figure 6 Query VI result

The result shows that in addition to Laura's account, the attacker had also successfully signed on to another account.

Since the attack began with phishing email, the next step is to identify how many employees were sent the phishing email. Following query was used for this investigation:

```
index=ir SenderAddress="hr-support@nexus-verify.thm"  
  
| stats count by recipient
```


Incident Detection and Analysis

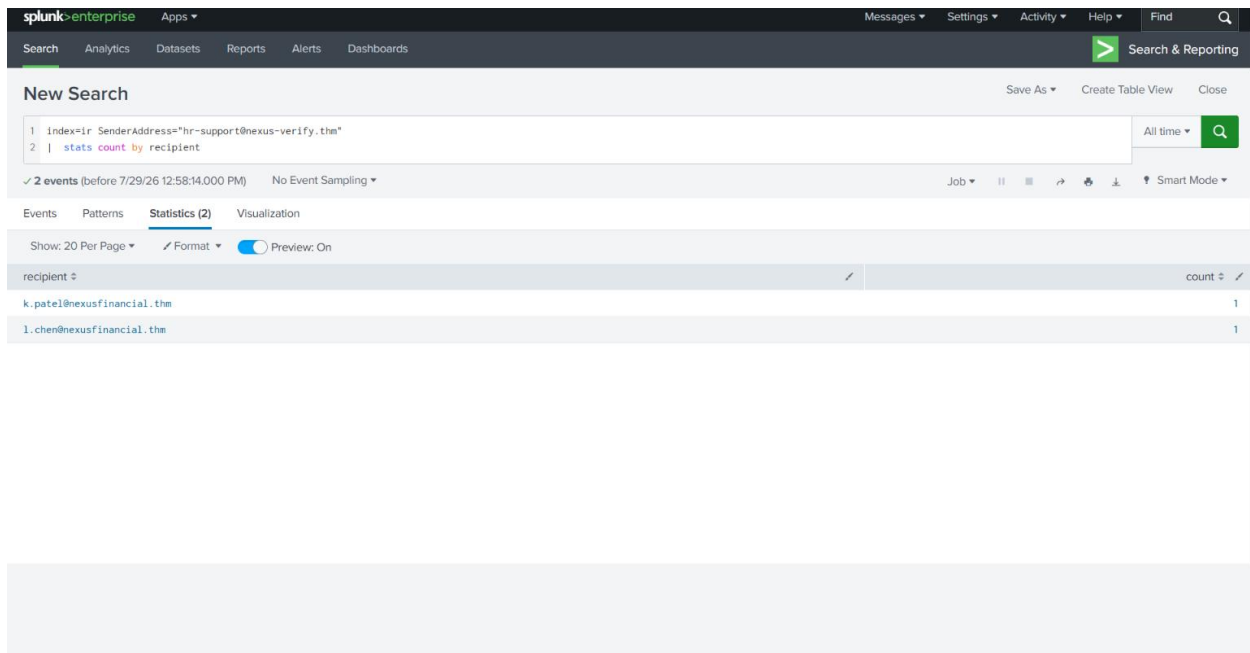


Figure 7 Query VII result

The result shows that the attacker sent phishing email to only two accounts and both of them were compromised.

The final investigation is to check if the attacker used inbox rules to cover their tracks. The following query was used:

```
index=ir Operation="New-InboxRule"
| table _time, user, Parameters{}.Value
```

Incident Detection and Analysis

The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk enterprise' logo and various menu items like 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'Search & Reporting' section with a 'New Search' button. The search query is: `1 index=ir Operation="New-InboxRule"`
`2 | table _time, user, Parameters().Value`. The results show 2 events. The first event is from 2026-03-30 16:58:48 for user l.chen@nexusfinancial.thm, with parameters: False, False, Junk Filter Update, security>alert;suspicious;password;verify, True, True. The second event is from 2026-03-30 17:02:40 for user k.patel@nexusfinancial.thm, with parameters: False, False, Security Updates, security>alert;password, True, True.

_time	user	Parameters().Value
2026-03-30 16:58:48	l.chen@nexusfinancial.thm	False False Junk Filter Update security>alert;suspicious;password;verify True True
2026-03-30 17:02:40	k.patel@nexusfinancial.thm	False False Security Updates security>alert;password True True

Figure 8 Query VIII result

The attacker used Junk Filter Update inbox rule in Laura's account and Security Update in the other compromised account.

Incident Detection and Analysis

3 Findings

1. The attacker (223.123.4.50) sent phishing email to two employees including Laura via the domain nexus-verify.thm.
2. The attacker's IP address originated from Amsterdam.
3. The attacker sent the email with the subject mentioning the change in HR policy and the notice required immediate action, creating a sense of urgency and fear.
4. The attacker used inbox rules to cover his tracks from Laura and the other victim.